

Paper number	Hamiltonian terms				Misc
1	H_{onsite} =onsite repulsion term for AAs occupying same space	H_{psc} =primary sequence constraint term	H_{pairwise} =pairwise interaction terms that represents favorable hydrophobic interactions between adjacent hydrophobic AAs		2D grid, encodes coordinates, not turns
2	E_{onsite} =penalizes configurations with overlaps among any two amino acids	E_{pw} =accounts for NN pairwise interaction energies among non-bonded AAs	E_{ext} = any external potentials other than the ones coming from interactions among the AAs defining the protein		In supplementary materials, 2D grid
3	E_{back} = penalizes the special case of overlaps of the chain folding onto itself	E_{overlap} = penalizes all possible overlaps, however less ancilla needed if E_{back} is introduced	E_{pair} = the energy given by the pairwise interaction of beads in the chain		
4	H_{back} =penalizes folding back onto itself	H_{redun} = penalizes occurrence of two redundant 3-bit strings (not a valid direction on the grid)	H_{olap} = penalizes folds in which any residues with i and j with $j>i+3$ occupy the same grid point, accompanying H_{back}	H_{pair} = interaction between non-bonded AAs that are adjacent on the lattice using HP or MJ interaction potential	Cubic lattice on 3D, treating H_{back} and H_{olap} separately reduces ancilla
5	H_{overlap} = to avoid self intersections	H_{pair} = account for the interaction between AAs			Cubic lattice on 3D, H_{overlap} taken from reference 4
6	They map Rosetta energy function to quantum, original paper https://pubs.acs.org/doi/10.1021/acs.jctc.7b00125				This is a peptide design paper, they use Rosetta energy function, they call this QPacker
7	H_{gc} =growth constraint term to avoid bifurcations	H_{ch} = chirality constraint to enforce correct stereochemistry of the side chains if present	H_{in} = interaction energies for MJ model		3D tetrahedral lattice
8	U_{coord} =unitary operation of coordinate calculations	U_f = oracle	U_{diff} = diffusion operator		Body-centered cubic lattice, HP model Grover to find the

					lowest energy
9	They use the Hamiltonian from reference 3				They don't really build a new Hamiltonian, they analyze annealing performance
10					Very similar to 8, Grover based, they detail the oracle building further in this paper
11	Oracle implements the energies, conducts required calculations to find and mark the answer states	E_{tot} =for every pair-wise interaction in the structure are summed using QC adders to find the total energy for each sequence of residues (states)	E_{th} =total energy is subtracted from a threshold energy value of the circuit for each state, and the ones with the negative result are the answer	Oracle negates the amplitude of the answer states, uncomputes all previous steps	Grover based
12	They use the Hamiltonian from the reference 7				They propose using a different ansatz (counterdiabatic -CD) and compare the performance among different quantum devices
13	$H = (\text{number of self crossings}) + \lambda \cdot (\text{distance between walk endpoints})$	After considering sampling SAW, they look at the Hamiltonian from 7 and propose a model where non-backtracking constraint is automatically enforced	Instead of using 4 qubits to describe the turns, they use binary encoding that uses 2 qubits. They define absolute and relative encoding of turns. Relative enforces the non-backtracking automatically		Starts with a simplified version of the problem; sampling self-avoiding walks on a 2d square lattice using QAOA
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